

Research Paper

The effects of the velocity distribution of impact generated water ice ejecta on exospheric escape from airless bodies

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ABSTRACT

We explore the velocity distribution of impact generated water ice ejecta from planetary surface. We show that the ejecta velocity distribution should be modeled by a mixed Maxwellian distribution, rather than the commonly used single Maxwellian distribution, in order to better capture the outcome of the different emission processes and the dynamics of the clustered water molecules in the ejecta. We find that, comparing to the mixed Maxwellian distribution, the single Maxwellian distribution may lead to a significant underestimation of the impact-ejected water molecules that stay gravitationally bound for exosphere models. While the relative difference is within 10% for celestial bodies with a large escape velocity such as Mercury, it can be about 1000% for celestial bodies with a small escape velocity such as Enceladus.

1. Introduction

Water ice is a fundamental compound throughout the solar system, and is present in many celestial bodies. Recent space missions have presented the existence of water ice on the Moon (Pieters et al., 2009; Li et al., 2018; Honniball et al., 2021; Colaprete et al., 2010; Schultz et al., 2010), Mercury (Lawrence et al., 2013), and Ceres (Prettyman et al., 2017). For icy moons outside the snowline, their surface are believed to be composed mostly of water ice. This paper concerns with the transport of the ejected water ice in the exosphere.

Farrell et al. (2013) proposed four mechanisms for the release of water molecules from the lunar regolith surface, including solar wind ion sputtering, electron stimulated desorption (ESD), photon stimulated desorption (PSD), and impact vaporization. Impact vaporization is believed to be the most important mechanism because it generates the largest source flux of polar water molecules. To simulate the transport of the water molecules by solar wind sputtering and impact vaporization, Farrell et al. (2015) used Monte Carlo simulation to estimates whether the water molecules will be surface bounded or escape based on the velocity distribution function. They assumed that the molecules ejected by sputtering follow a Sigmund–Thompson distribution (Falcone, 1987), while those ejected by impact vaporization generally follow a Maxwellian distribution with a temperature estimated by Cintala (1992). A detailed review of the transport of water

molecules in the exosphere can be found in Schörghofer et al. (2021) and reference therein.

Experimental studies (Kato et al., 1995; Buhl et al., 2014; Arakawa, 1999) have showed that clustering of water molecules are commonly found during hypervelocity ice impact. Such experiments are motivated by modeling the secondary craters on planetary surfaces caused by the impact ejecta. The ejecta considered are relatively large grains ($>0.1 \mu\text{m}$). Fragments produced after a hypervelocity impact event are found to follow a power-law distribution in both size and mass. Furthermore, clustering is also found to be significant at a much smaller size scale (molecular level), which are typically considered in exosphere transport modeling. Ulibarri (2022) and Nelson (2016) used a time of flight mass spectrometer to detect the fragmentation of water ice upon the hypervelocity impact of an iron grain. These studies examined impact velocities ranging up to 25 km/s (Nelson, 2016) and demonstrated the presence of large clusters. For instance, large clusters up to $(\text{H}_2\text{O})_{33}\text{H}$ were found for impact velocity at 16 km/s (Ulibarri, 2022). These findings provide clear evidence of the existence of large water clusters generated during hypervelocity impact in a wide range of velocities.

The mechanism underlying the fragmentation of ice during hypervelocity impact is rooted in the hydrogen bonding between water

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molecules in an ice structure. Depending on the energy released by the impact, some of these bonds can survive and link individual molecules, leading to the creation of water clusters. These clusters can have a much greater mass than individual water molecules and, therefore, a lower average velocity. The surface bounded exosphere is directly determined by the velocity distribution of the ejected molecules. The released water molecules in the form of clusters during hypervelocity impact can lead to an underestimation of the proportion of ice captured by gravity if one assume that most water molecules are released as single molecules. Therefore an understanding of the formation of water produced during impact is critically important to model the surface-bounded exosphere for ice-rich regions of planetary bodies.

Previous experiments mostly focused on measuring the size distribution of fragments (Kato et al., 1995; Buhl et al., 2014). However, the dynamics of the ejecta has not been addressed extensively. One of the primary challenges in studying the ejecta dynamics is the wide range of fragment sizes and velocities that can be generated during an impact. Smaller fragments tend to have higher velocities, making them more challenging to track with conventional detectors. Additionally, the angular distribution of fragments can be highly anisotropic as they can be ejected in different directions with different velocities (Bernardini et al., 2019). This anisotropy can make it difficult to obtain a complete measurement of the ejecta dynamics. Huang et al. (2023) recently investigated the fragmentation of an ice nano cluster upon hypervelocity impact using reactive molecular dynamics simulation. The combination of bond calculation based on first-principle trained force field and Newtonian equations provide a way to track both chemical bond breaking and forming as well as the dynamics of molecules. A similar approach has also been used to model chemical reactions that leads to the formation of water molecules on the lunar surface including the dynamics of those molecules (Huang et al., 2021, 2022).

In this study, we extend the analysis in Farrell et al. (2013) by performing an analytical study of the surface bounded exosphere to take into account of the clustering of water molecules during impact. We investigate how clusters of ice affect the portion of water molecules captured by gravity, which is crucial for understanding the distribution and transport of water molecules on icy bodies. We test our analytical model on several planetary bodies with ice regions, including the Moon, Mercury, and Ceres where the presence of water ice has been confirmed by various missions. By considering the clustering of water molecules and the mechanisms responsible for the transport of water molecules during impact, we address the geophysical characteristics of these icy bodies.

2. Method and formulation

We use the single Maxwellian velocity distribution function (VDF) to model the ejecta that contains only individual water molecules, and the mixed Maxwellian VDF to take into account of the effect of clustering. The commonly used single Maxwellian VDF is:

$$f(v) = \left(\frac{m}{2\pi kT}\right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}} \quad (1)$$

As discussed before, the power-law distribution provides the best fit for the size distribution of fragments (Kato et al., 1995 and reference within). We denote S as the size of the cluster, i.e. $(\text{H}_2\text{O})_S$. The cumulative size distribution function (CDF) of the water clusters is:

$$CDF(> S) \propto S^{-a} \quad (2)$$

where the average power index a is 0.69 (Kato et al., 1995). One can then derive the size distribution based on the cumulative distribution function as:

$$n(S) \propto S^{-a-1} \quad (3)$$

The proportion of each cluster from cluster size 1 to S is then:

$$R_{\text{cluster}}(S) = \frac{S^{-a-1}}{\sum_{S=1}^{S_{\text{cutoff}}} S^{-a-1}} \quad (4)$$

As a cluster contains S water molecules, the probability of finding a water molecule in a cluster with size S is then:

$$R_{\text{H}_2\text{O}}(S) = S \times \frac{S^{-a-1}}{\sum_{S=1}^{S_{\text{cutoff}}} S \times S^{-a-1}} = \frac{S^{-a}}{\sum_{S=1}^{S_{\text{cutoff}}} S^{-a}} \quad (5)$$

The velocity distribution for different sizes are also different because of the difference in mass. The VDF for cluster with size of S is:

$$f_S(v) = \left(\frac{Sm}{2\pi kT}\right)^{3/2} 4\pi v^2 e^{-\frac{Smv^2}{2kT}} \quad (6)$$

Taking the proportion of size as the weight for each specific distribution, the overall VDF is then in the form of:

$$f_{\text{all}}(v) = \sum_{S=1}^{S_{\text{cutoff}}} \frac{S^{-a}}{\sum_{S=1}^{S_{\text{cutoff}}} S^{-a}} \left(\frac{Sm}{2\pi kT}\right)^{3/2} 4\pi v^2 e^{-\frac{Smv^2}{2kT}} \quad (7)$$

By setting the escape velocity for a specific celestial body, one can then estimate how much water can be captured by gravity. Numerical tests show that $S_{\text{cutoff}} = 100$ will result in a variation of <1% of the results. Hence, S_{cutoff} is taken as 100 so that $R_{\text{H}_2\text{O}}$ is not sensitive to the choice of S_{cutoff} .

3. Results and discussion

3.1. Dependence on the escape velocity

Fig. 1(a) compares the single Maxwellian VDF (dashed line) and mixed Maxwellian VDF (solid line) for different vaporization temperatures. The range of the impact velocities considered in this study covers a temperature range from $T = 1000$ K to $T = 10000$ K, based on previous research on impact velocity distribution (Cintala, 1992) and temperature of water vapor under the compression of shock (Lyzena et al., 1982). Each curve is color-coded by the vaporization temperature. The mixed Maxwellian VDFs exhibit a lower average velocity and a narrower velocity distribution compared to the single Maxwellian distributions. This difference can be attributed to the presence of larger particles, or clusters, which have a higher equivalent mass than smaller particles. The thick black curves indicate the escape velocity for several celestial bodies, including Europa, the Moon, and Mercury. Integrating the VDFs in Fig. 1(a), one can obtain the cumulative distribution function (CDF) in Fig. 1(b). The intersection points between each CDF curves and the escape velocity dashed curves shows the probability of finding an ejected particle with a velocity lower than the escape velocity, denoted by $P(V < V_{\text{escape}})$. This value is an important metric for understanding the likelihood of ejecta deposition and whether it can escape the gravity or not. The mixed Maxwellian distribution tends to exhibit a higher probability of finding ejected particles with velocities below the escape velocity, whereas the single Maxwellian distribution yields a lower probability.

We define $\omega = \frac{N_{\text{bounded}}}{N_{\text{all}}}$ as the ratio of the number of molecules bounded by gravity N_{bounded} to that of the total water molecules ejected N_{all} . We use the subscript “single” to denote results from using the single Maxwellian VDF and the subscript “mixed” to denote that from using the mixed Maxwellian VDF. Fig. 2(a) compares ω for single and mixed Maxwellian VDF, and Fig. 2(b) compares $\omega_{\text{mixed}} - \omega_{\text{single}}$ for Mercury, Europa, and the Moon. The figure provides a clear illustration of the variation in the difference between ω_{mixed} and ω_{single} across the three celestial bodies with different escape velocities. This variation indicates the importance of the escape velocity in determining whether a single or mixed Maxwellian distribution should be used to model the velocity distribution of water molecules. As the impact vaporization temperature becomes higher, the difference increases significantly. Depending on the escape velocity, the difference between the results using the two VDFs may start to decrease at extremely high vaporization temperatures. Fig. 2(b) shows that, for Europa, the maximum difference occurs at a temperature of about 6000 K. This is because an extremely high vaporization temperatures can generate a greater proportion of

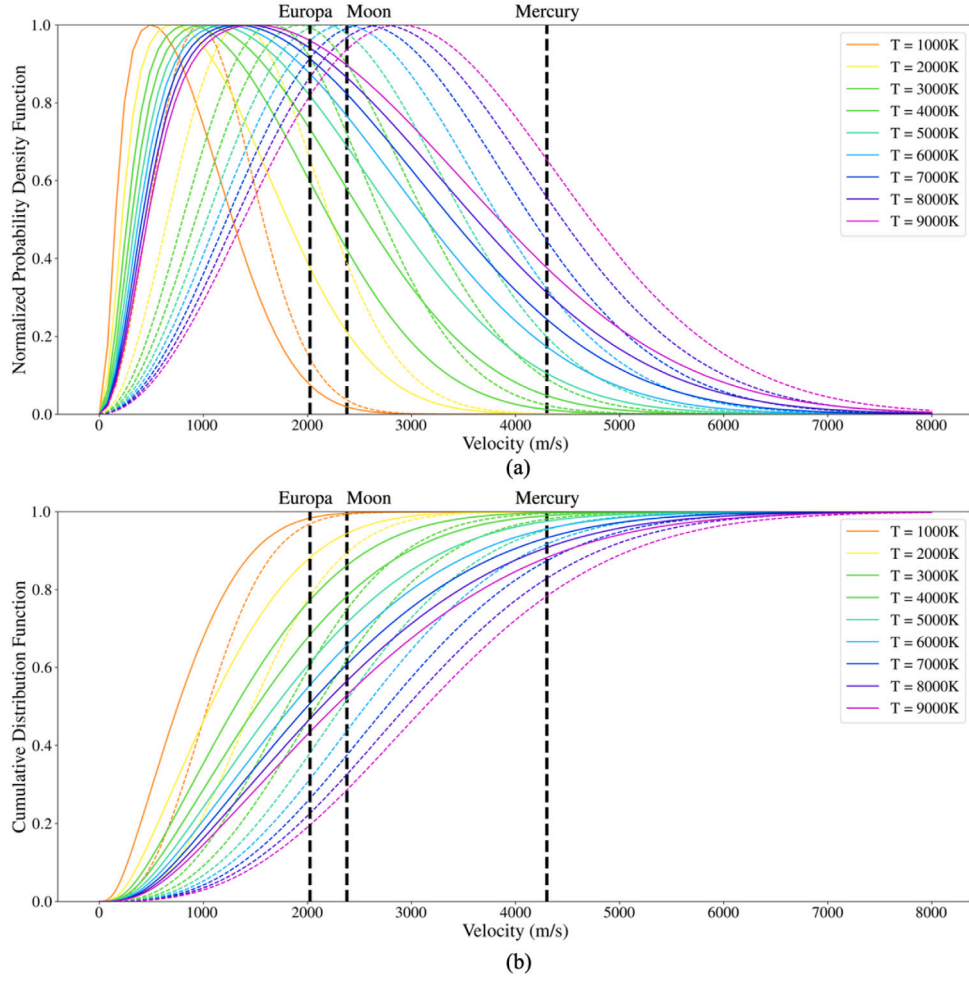


Fig. 1. (a) Velocity distribution function, (b) cumulative distribution function for single Maxwellian (dashed curves) and mixed Maxwellian (solid curves). black dashed line indicates the escape velocity of Europa, the Moon and Mercury. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

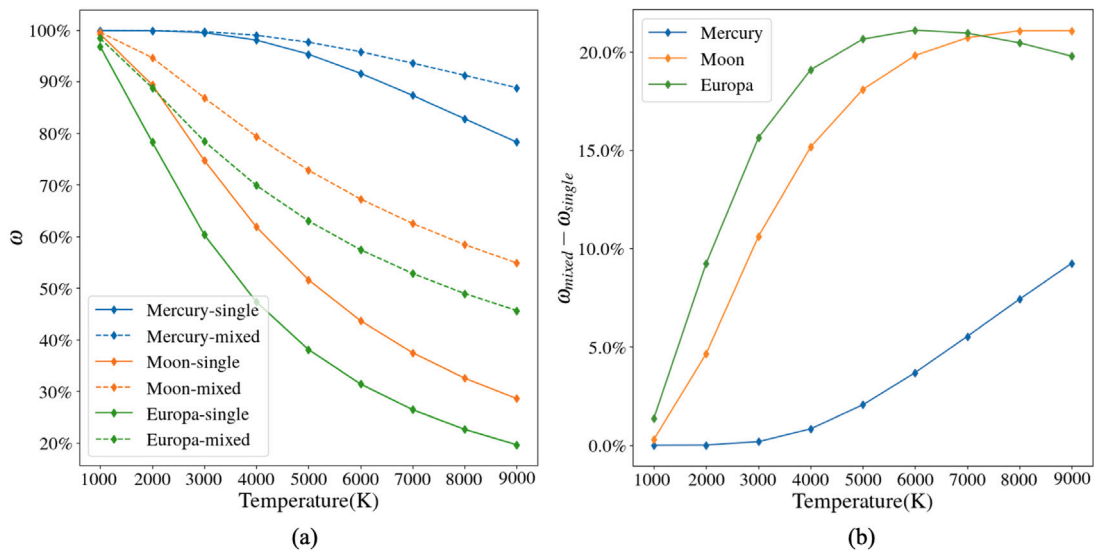


Fig. 2. Comparison of (a) surface bounded portion between single Maxwellian distribution and mixed Maxwellian distribution and (b) the difference of surface bounded portion between single Maxwellian distribution and mixed Maxwellian distribution for Mercury, Europa, and the Moon.

the ejected particles with velocities above the escape velocity. As a result, fewer water molecules in either the single Maxwellian VDF or the mixed Maxwellian VDF fall within the range of the escape velocity.

We further calculate the relative difference, ϵ , between the results using the mixed Maxwellian VDF and the single Maxwellian VDF:

$$\epsilon = \frac{\omega_{\text{mixed}} - \omega_{\text{single}}}{\omega_{\text{single}}} \quad (8)$$

The results are shown in Fig. 3. We find the relative difference is strongly affected by the escape velocity of planetary bodies. For Mercury, where the escape velocity is $V_{\text{escape}} = 4300$ m/s, the largest difference for the impact temperature range considered is within 10%, suggesting that a single Maxwellian VDF may be sufficient to accurately model the velocity distribution of water molecules in this case. However, as the escape velocity decreases, the relative difference increases significantly. For the Moon, the largest relative difference for the impact temperature range considered is near 80%. For Europa, the relative difference is near 120%. The relative difference is even more significant for smaller planetary bodies, such as Ceres, Tethys, and Enceladus. For instance, for Enceladus, the relative difference for the impact temperature range considered is as high as about 1000%. These results highlight the limitations of using a single Maxwellian distribution in modeling the velocity distribution of water molecules in low escape velocity environments.

3.2. Sensitivity test of power index

The significant difference observed in the case of Europa highlights the necessity of considering the effects of the cluster distributions when modeling the exosphere of icy moons. Many icy moons are believed to harbor organic molecules, which could have important astrobiological implications. In this section, we investigated the sensitivity of the selection of the power index, a , in Eq. (2). The power index determines the distribution of clusters, which is an essential component of the analysis presented in this paper. By varying the value of a , we were able to test the robustness of our results to changes in the cluster distribution. Our analysis reveals that the choice of power index can have a significant impact on the relative difference defined in Eq. (8), particularly in scenarios with low escape velocities. Specifically, we find that when the power index is increased, the relative abundance of surface bounded water molecules decreases as there are fewer clusters.

Based on previous studies, such as Kato et al. (1995), the power index in Eq. (2) is expected to fall within the range of 0.67 to 0.72. To test the sensitivity of our results to changes in the power index, we selected the upper bound of 0.72 and the lower bound of 0.67 as the extreme values for our analysis. Our analysis reveals that the deviation in the relative difference for Mercury, Europa, and the Moon is within about 2%, 10%, and 20%, respectively. This suggests that the selection of the power index is valid and that our results are robust to changes in the cluster distribution for these planetary bodies. However, for smaller planetary bodies, such as Ceres, Tethys, and Enceladus, we find the deviation in the relative difference can be as high as 200%. This suggests that a more accurate estimation of the power-law index is needed for smaller bodies. In addition, the change in the power-index due to molecular structure and possible chemical reactions will also need to be considered.

3.3. A general model for all exosphere species

The model presented in this paper has broad applicability beyond the study of water molecules in hypervelocity impact. It is reasonable to assume that our findings could be extended to other molecules or atoms commonly found in the exosphere that include clusters. For instance, Killen et al. (2005) proposed that the calcium in the exosphere of Mercury could be a combination of both Ca atoms and Ca-Ca dimers. The Ca-Ca dimers are thought to be produced by the vaporization of

larger Ca clusters. This could also indicate that larger Ca clusters are also possible to exist in Mercury's exosphere at a lower altitude as the scale height of those larger cluster would be smaller:

$$H_S = \frac{kT}{m_S g} = \frac{kT}{S m_1 g} \quad (9)$$

where m_S is the mass of $(\text{Ca})_S$, and $S = 1$ corresponds to the single Ca atom case. From a global point of view, the overall exosphere is composed of multiple layers with scale height of $(\text{Ca})_S$ equaling to H_S .

In general, whether it is necessary to use the mixed Maxwellian VDF or the single Maxwellian VDF depends on the species of interest. If there is no or very weak binding energy between the atoms or molecules, then the ejecta would mostly consist of individual atoms or molecules. For volatile species such as water ice, Nelson (2016) tested the impact of iron to thick water substrate for the velocity from 0–25+ km/s. The range covers average impact velocity of most celestial bodies which provides a good reference for the validation studies here. While the mechanisms underlying fragmentation can be complex and multifaceted, the findings of these experiments suggest that clustering and fragmentation are likely to occur under certain conditions, particularly at higher impact velocities. For mineral species, there have been extensive discussions regarding the vaporization process, in particular, complete vaporization and incomplete vaporization (Killen et al. (2005), Killen and Hahn (2015) and Killen (2016) and references therein). Depending on the impact velocity, when the pressure and temperature created by the impact leads to incomplete vaporization, it is also necessary to consider the mixed Maxwellian distribution discussed in this paper.

4. Conclusions

This paper proposed a framework to analyze the transport of molecules in exospheres, particularly those composed of clustered molecules or atoms, and to establish new exosphere models with the clustering taken into account. In this study, we construct a mixed Maxwellian VDF by taking the power-law distribution as the size distribution of impact ejecta and calculated the weighted distribution. The resulting VDF behaves differently from the single Maxwellian VDF. Depending on the escape velocity, the amount of water captured by gravity can be significantly underestimated. We quantitatively compared the relative difference of surface bounded proportion between the two assumptions. For planets with a large escape velocity such as Mercury, the relative difference is about 10%. However, for smaller icy bodies such as Europa, the relative difference is about 120%. For even smaller bodies such as Enceladus, the relative difference can be about 1000%. The results presented highlight the need to accurately model the VDF of the ejected water molecules in order to fully understand the physical processes that govern the transport of water molecules in the exosphere.

The analysis of this paper applies the power law index provided in Kato et al. (1995). We find that it is important to carefully select the power index in order to accurately model the behavior of water molecules and ice particles in different space environments. For instance, with a small variation of 8% in the power law index, the resulting relative difference can be as high as 200%. Future research should carry out a comprehensive analysis of the power index by combining theoretical modeling, experimental studies, and *in-situ* measurement. Future missions such as the Europa Clipper mission (Kempf et al., 2012) will provide a more accurate *in-situ* measurement of the power index through mass spectrometer measurement of ice grain hypervelocity impact.

The temperature range used in this study provides a general benchmark for planetary impacts. The exact surface temperature at the impact site varies depending on the properties of the projectile and target as well as the impact velocity. Future study will need to carefully characterize the temperature range used in the model. Moreover,

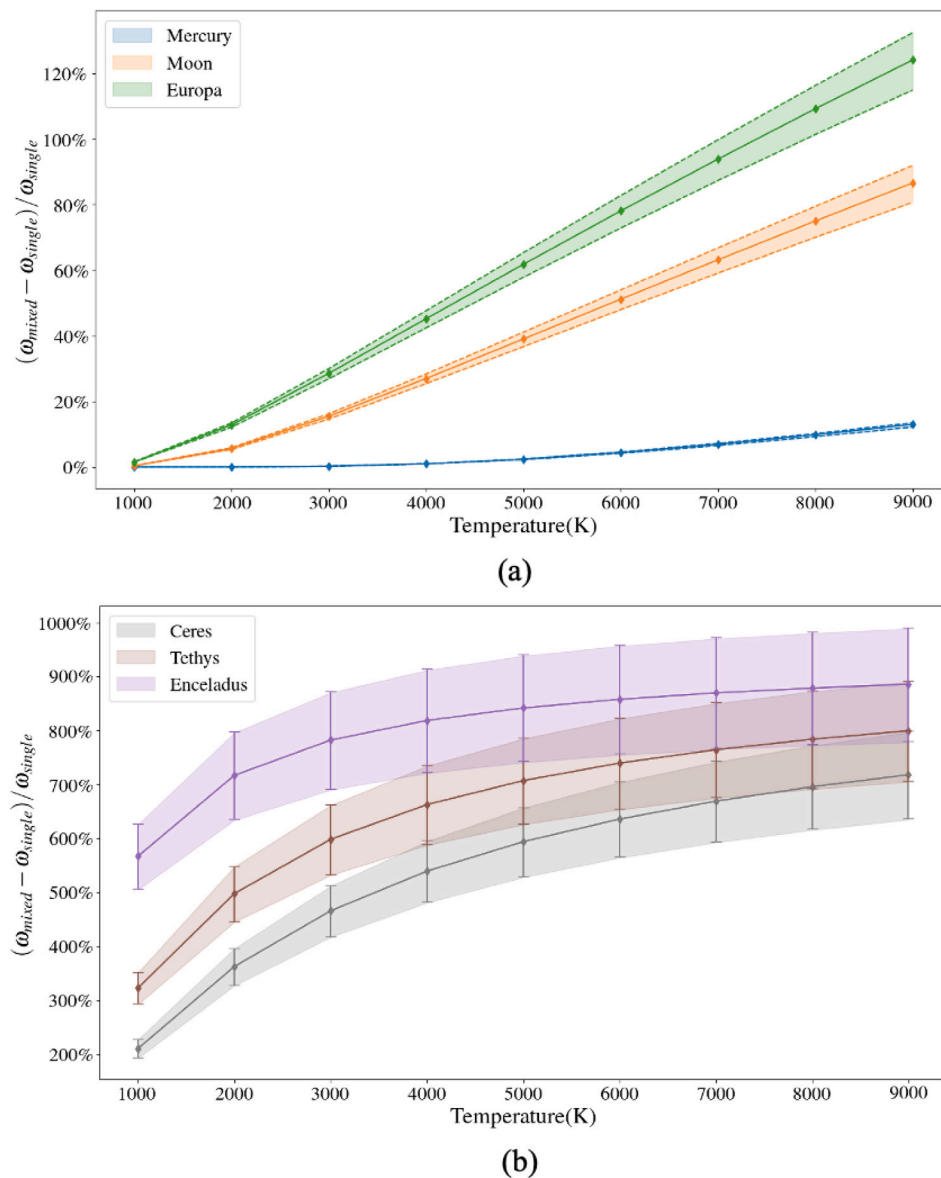


Fig. 3. Comparison of the relative difference of surface bounded portion between single Maxwellian distribution and mixed Maxwellian distribution for (a) Mercury, Moon, Europa and (b) Ceres, Tethys and Enceladus.

previous studies mostly assumed the role of impact vaporization to be less important compared to sputtering and sublimation for the Galilean moons (Plainaki et al., 2010). However, our results suggest that ignoring the contribution of impact vaporization may lead to a significant underestimation of the amount of water in the exosphere for these bodies. Future missions such as the exploration of Ganymede by Jupiter Icy Moons Explorer (JUICE) (Grasset et al., 2013) will provide more insights into micrometeorite impact on the icy moons. Detailed modeling studies of hypervelocity ice impact, such as that based on molecular dynamics simulation (Huang et al., 2021), are also needed in order to more accurately characterize the vaporization temperature, and to provide further validation for the use of the mixed-Maxwellian distribution. While the analysis presented focuses on water ice ejecta, the model is also applicable to the molecules and atoms of other volatiles and refractory materials in the exosphere, such as calcium (Killen et al., 2005) and calcium-oxides (Killen, 2016). Future studies will also need to extend the model to other species detected in the exosphere and compare with in-situ measurement of the exosphere structure.

CRediT authorship contribution statement

Ziyu Huang: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Liam S. Morrissey:** Conceptualization, Investigation, Methodology. **Orenthal J. Tucker:** Conceptualization, Investigation, Methodology. **Ken-ichi Nomura:** Conceptualization, Investigation, Methodology, Software. **Aiichiro Nakano:** Conceptualization, Investigation, Methodology, Software. **Joseph Wang:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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